Unit-I

Risk Management & Delivery management

Risk Management in Software Engineering

- **Definition of Risk:**
  - Risks in software engineering refer to potential future events that can negatively impact a project's success.
- **Types of Risks:**
  - **Project Risks:** Budget overruns, missed deadlines.
  - **Technical Risks:** Technology failures, poor design.
  - **Business Risks:** Market changes, poor product fit.

Risk Management

Uncertainty and loss are the two constants of risk. Identify it, project it, refine it, and plan for it — before it happens, not after.

The Risk Management Process

- **Identify Risks:**
    - Techniques: Brainstorming, checklists, expert interviews.
- **Analyze Risks:**
    - Assess probability and impact.
- **Plan Risk Mitigation:**
    - **Strategies: Avoidance, mitigation, acceptance, transfer.**
- **Monitor and Control Risks:**
    - Continuous monitoring and adjusting plans as needed.

Reactive vs. Proactive Risk Strategies

REACTIVE (“Indiana Jones” strategy)

- The team does nothing about risks until something goes wrong
- “Don't worry, I'll think of something!” — problems handled heroically, in the moment
- At best, resources are set aside “just in case”
- Failure escalates into fire-fighting, then crisis management

PROACTIVE (recommended)

- Begins long before technical work starts
- Potential risks are identified, probability and impact are assessed, and risks are ranked
- The team builds a plan for managing risk — avoidance first, then contingency
- Enables a controlled, effective response rather than a scramble

Categories of Software Risk

Project risks

Threaten the project plan: budget, schedule, personnel, resource, stakeholder and requirements problems.

Technical risks

Threaten quality and timeliness: design, implementation, interface, verification, maintenance, or “leading-edge” technology problems.

Business risks

Threaten the viability of the product: market, strategic, sales, management and budget risk.

By predictability (Charette)

Known risks

Can be uncovered after careful evaluation of the project plan and environment (e.g. unrealistic delivery date).

Predictable risks

Extrapolated from past project experience (e.g. staff turnover, poor customer communication).

Unpredictable risks

The “joker in the deck” — can occur, but are extremely difficult to identify in advance.

Risk Projection: Building a Risk Table

Risk	Category	Probability	Impact
Size estimate may be significantly low	PS	60%	2
Larger number of users than planned	PS	30%	3
Less reuse than planned	PS	70%	2
End-users resist the system	BU	40%	3
Delivery deadline will be tightened	BU	50%	2
Customer will change requirements	PS	80%	2
Lack of training on tools	DE	80%	3
Staff turnover will be high	ST	60%	2

IMPACT SCALE

1. Catastrophic
2. Critical
3. Marginal
4. Negligible

Categories: PS = size, BU = business, DE = development env., ST = staff

Sort the table by probability and impact. High-probability, high-impact risks rise to the top; draw a cutoff line — only risks above it get full RMMM treatment. Revisit the table at regular intervals as the project proceeds.

Risk Impact & Risk Exposure :

How much money (or loss) is expected because of a particular risk?

$$RE = P \times C$$

P = probability of occurrence · C = cost to the project if the risk occurs

WORKED EXAMPLE — REUSE RISK

Risk identification: Only 70% of components scheduled for reuse will actually be integrated; the rest must be custom developed.

Risk probability: 80% (likely).

Risk impact: 60 reusable components were planned; 30% (18 components) must be built from scratch. At 100 LOC/component and $14.00/LOC → cost = 18 × 100 × 14 = $25,200.

Risk exposure: RE = 0.80 × 25,200 ≈ $20,200.

Rule of thumb: if total RE across all risks exceeds ~50% of the project cost estimate, the viability of the project itself must be re-evaluated.

Risk Refinement & the RMMM Plan

REFINEMENT: CTC FORMAT

(Condition–Transition–Consequence)

“Given that <condition>, then there is concern that (possibly) <consequence>.”

E.g. Given that reusable components must meet a design standard some do not meet, there is concern that only 70% may be integrated as-is.

RMMM: THREE-PART STRATEGY

- **Mitigation.** Avoidance - address root causes before the risk becomes real.
- **Monitoring.** Track factors that signal the risk is becoming more or less likely.
- **Management.** Contingency plan for when mitigation fails and the risk occurs.

RISK INFORMATION SHEET — EXAMPLE

Risk ID / Prob. / Impact: P02-4-32 · 80% · High

Description: Only 70% of components scheduled for reuse will be integrated; remainder custom-built.

Mitigation/monitoring: Contact third parties on design-standard conformance; press for interface-standard completion; check language support.

Management/contingency: RE = $20,200, allocated to project contingency cost; revise schedule assuming 18 components are custom-built.

Project Delivery Management

- Bringing estimation, scheduling and risk management together into a disciplined, repeatable path to on-time, on-budget delivery.

The W5HH Principle (Boehm)

"An organizing principle that scales down to provide simple plans for simple projects" — seven questions that define a project's key characteristics.

Why is the system being developed?
Validate the business reason for the work.

What will be done?
Define the task set required for the project.

When will it be done?
Build the schedule: tasks and milestones over time.

Who is responsible for a function?
Define each team member's role & responsibility.

Where are they organizationally located?
Customers, users and other stakeholders too.

How will it be done?
Define the management and technical strategy - technically and managerially.

How much of each resource is needed?
Derived from estimates built on the answers above.

Critical Practices for On-Time Delivery

The Airlie Council's practices, consistently used by high-performing software organisations:

1. **Metric-based project management**
   Decisions grounded in measured data, not gut feel.
2. **Empirical cost & schedule estimation**
   Decomposition + calibrated empirical models, cross-checked.
3. **Earned value tracking**
   Quantitative, early-warning progress measurement.
4. **Defect tracking against quality targets**
   Visibility into quality trends throughout the process.
5. **People-aware management**
   Recognise that motivated, well-supported teams deliver.

Integration: Estimation + Scheduling + Risk

Estimate
Decomposition + empirical models → size, cost, effort

Schedule
Task network → Gantt chart → tracking & earned value

Manage Risk
Identify → project → refine → RMMM plan

Deliver
W5HH + critical practices → on-time, on-budget release

Each stage feeds the next, and each feeds back: a slipping schedule triggers a risk re-assessment; a new risk reshapes the estimate; a revised estimate resets the schedule.

THE MANAGER'S JOB

Keep the estimate honest, the schedule visible, and the risks in front of the team — not behind it. Delivery is not a single event at the end of the project; it is the accumulated discipline of every week that preceded it.